

The City Called Cell

I PUC · Ch 8 · Cell: The Unit of Life

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



Welcome to Cell City

"Every living thing is made of cells," said Prof. BAIO. **Schleiden** said it for plants (1838) and **Schwann** for animals (1839) — the **cell theory**. Later **Virchow** added, "Every cell comes from a cell that already exists."

💡 **REMEMBER**

Virchow (1855): Omnis cellula-e cellula — new cells arise from pre-existing cells.

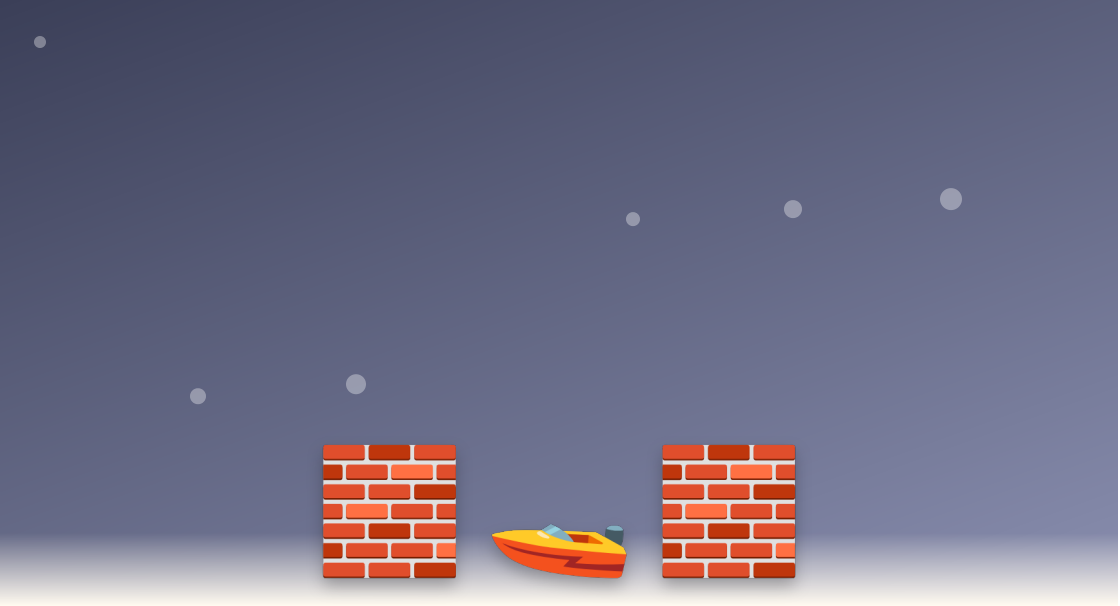


PAGE 2 OF 8

Villages and Cities

Bacteria lived in simple villages — **prokaryotic cells** with no nuclear membrane, only a **nucleoid**, small rings of DNA called **plasmids** and folds called **mesosomes**. Plants and animals lived in big cities — **eukaryotic cells** with a true nucleus and many organelles.

> 💡 **REMEMBER**
> **Prokaryotes have 70S ribosomes and no membrane-bound organelles.**



The City Wall

Around every city ran the **plasma membrane**. **Singer and Nicolson** (1972) described it as the **fluid mosaic model**: a sea of lipids, two layers thick, with proteins floating in it like boats. Plant cities had an extra outer wall of cellulose — the **cell wall**.

> 💡 **REMEMBER**
> **The membrane is selectively permeable; its lipids are quasi-fluid.**



Roads and Post Office

The **endoplasmic reticulum** was the road network. **Rough ER**, dotted with ribosome workers, made proteins; **smooth ER** made lipids. The **Golgi apparatus**, with its stack of flat **cisternae**, was the post office — packing goods and sending them out.

💡 **REMEMBER**

Golgi has a cis (forming) and a trans (maturing) face; it forms glycoproteins and glycolipids.



Recycling and Storage

Lysosomes were recycling bins full of hydrolytic enzymes — the "suicide bags". **Vacuoles**, bound by the **tonoplast**, stored sap; in plant cells one big vacuole could fill 90% of the space!

💡 **REMEMBER**

Lysosomal enzymes work best at an acidic pH.



PAGE 6 OF 8

The Power Station

Mitochondria were the power stations, their inner walls folded into **cristae**, making ATP by aerobic respiration. They even had their own circular DNA and 70S ribosomes — that is why they are called **semi-autonomous**.

💡 REMEMBER

Mitochondria: double membrane, cristae, matrix with DNA, RNA and 70S ribosomes.

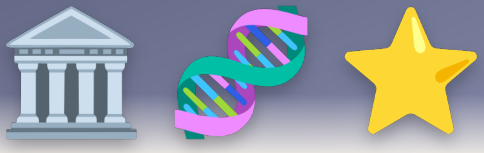


The Solar Farm

Plant cities had **plastids**. Green **chloroplasts** trapped sunlight in stacks of **thylakoids** called **grana**. Colourful **chromoplasts** painted fruits and flowers, and colourless **leucoplasts** stored food — starch, oil or protein.

💡 **REMEMBER**

Amyloplasts store starch, elaioplasts store oils, aleuroplasts store proteins.



PAGE 8 OF 8

City Hall

At the centre stood the **nucleus**, wrapped in a double membrane with pores. Inside, the **nucleolus** made rRNA and the **chromatin** held the DNA. Nearby, the **centrosome** with its two **centrioles** got ready for cell division.

💡 REMEMBER

Nucleolus — site of rRNA synthesis · Centrioles lie at right angles to each other.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

Schleiden p.1 · Schwann p.1 · cell theory p.1 · Virchow p.1 · prokaryotic cells p.2 · nucleoid p.2 · plasmids p.2 · mesosomes p.2 · eukaryotic cells p.2 · plasma membrane p.3 · Singer and Nicolson p.3 · fluid mosaic model p.3 · cell wall p.3 · endoplasmic reticulum p.4 · Rough ER p.4 · smooth ER p.4 · Golgi apparatus p.4 · cisternae p.4 · Lysosomes p.5 · Vacuoles p.5 · tonoplast p.5 · Mitochondria p.6 · cristae p.6 · semi-autonomous p.6 · plastids p.7 · chloroplasts p.7 · thylakoids p.7 · grana p.7 · chromoplasts p.7 · leucoplasts p.7 · nucleus p.8 · nucleolus p.8 · chromatin p.8 · centrosome p.8 · centrioles p.8



QUICK QUIZ

Do you remember?

1. "Omnis cellula-e cellula" was said by:

- Schleiden
- Schwann
- Virchow
- Robert Brown

2. The fluid mosaic model was given by:

- Watson and Crick
- Singer and Nicolson
- Schleiden and Schwann
- Jacob and Monod

3. Which organelle is called the suicide bag?

- Lysosome
- Golgi body
- Vacuole
- Ribosome

4. rRNA is made in the:

- Nucleolus
- Mitochondrion
- Golgi body
- Centriole

5. Mesosomes are found in:

- Plant cells
- Animal cells
- Prokaryotic cells
- Fungal cells

Answers: 1-Virchow · 2-Singer and Nicolson · 3-Lysosome · 4-Nucleolus · 5-Prokaryotic cells